A Spatio-Temporal Hierarchical Diffusion Framework for Training-Free Perceptual Remote Sensing Image Compression

Yangxuan Cheng, Fanyang Meng, Hao Qi, Han Shen, Zhongqiang Zhang, Ye Wang, *Member, IEEE*, and Yongsheng Liang, *Member, IEEE*

Abstract*—Remote sensing (RS) data contains vast amounts of geographic structural information and fine-grained textural details, making compression essential and necessitating efficient processing for downstream interpretation tasks. Generative models have shown great potential in learned image compression (LIC), offering high compression rates and exceptional generation quality. Among these, diffusion models stand out for their ability to produce high-quality reconstructions. However, they may hallucinate small structures and textures due to significant information loss at limited bitrates. Furthermore, compression based on the diffusion process suffers from long inference times caused by numerous denoising steps, which hinders practical deployment. We propose spatio-temporal hierarchical codebook diffusion (STHCD), a training-free framework that dynamically allocates resources to align with this hierarchy. STHCD introduces three synergistic components: 1) spatial residual transformation via parameter-free unshuffle operations to adapt spatial granularity; 2) channel-wise codebook splitting for flexible bit allocation without memory explosion; and 3) a time-dependent hierarchical policy to orchestrate these operations across timesteps. Extensive experiments on nine datasets show that STHCD significantly outperforms state-of-the-art methods in perceptual metrics [learned perceptual image patch similarity (LPIPS), deep image structure and texture similarity (DISTS), Fréchet inception distance (FID)]. At equivalent bitrates [~0.12 bits per pixel (bpp)], STHCD achieves an 8× ~ 12.5× speedup (80 ~ 125 versus 1000 steps) over standard temporal codebook diffusion (TCD) while optimizing LPIPS by 29.5%. Furthermore, STHCD can achieve a remarkable 15-step diffusion at the same bitrate, with only a minor LPIPS drop, establishing a new paradigm for hierarchical RS compression. Our project is available at https://sthcd.github.io

Received 4 December 2025; revised 1 February 2026; accepted 16 March 2026. Date of publication 19 March 2026; date of current version 26 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62571160, Grant 62031013, and Grant 62501333; in part by the Major Key Project of Pengcheng Laboratory under Grant PCL2024A01 and Grant PCL2025A03; and in part by the Engineering Technology Research and Development Center of Guangdong Provincial Universities under Grant 2024GCZX004. *(Corresponding author: Yongsheng Liang.)*

Yangxuan Cheng, Hao Qi, and Han Shen are with the School of Information Science and Technology, Harbin Institute of Technology, Shenzhen 518055, China, and also with the Pengcheng Laboratory, Shenzhen 518060, China (e-mail: chengyx@stu.hit.edu.cn; qih@pcl.ac.cn; shenh@stu.hit.edu.cn).

Fanyang Meng, Zhongqiang Zhang, and Ye Wang are with the Pengcheng Laboratory, Shenzhen 518060, China (e-mail: mengfy@pcl.ac.cn; zhangzhq@pcl.ac.cn; wangy02@pcl.ac.cn).

Yongsheng Liang is with the School of Information Science and Technology, Harbin Institute of Technology, Shenzhen 518055, China, and also with the College of Big Data and Internet, Shenzhen Technology University, Shenzhen 518118, China (e-mail: liangys@hit.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2026.3675642, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2026.3675642

***Index Terms*—Diffusion models, generative models, remote sensing image compression (RSIC).**

NOMENCLATURE

$\mathbf{x}_0$	Input image.
$\mathbf{z}_0$	Latent representation of the input image.
$\mathbf{z}_t$	Noisy latent variable at timestep t.
$\hat{\mathbf{z}}_{0\|t}$	Predicted clean latent from $\mathbf{z}_t$.
$\mathcal{E}/\mathcal{D}$	VAE encoder/decoder.
ϵ_θ	Noise prediction network.
$\alpha_t, \bar{\alpha}_t$	Noise schedule parameters.
σ_t	Standard deviation of noise at timestep t.
$\mathbf{c}_{t-1}$	Pre-sampled codebook at timestep t.
$\mathbf{c}_{t-1}^{(k)}$	kth codebook vector in $\mathbf{c}_t$.
k_t	Index of the selected codebook vector at timestep t.
$\downarrow_s$ / $\uparrow_s$	Residual (un)shuffle operation with factor s.
Q_n	Codebook splitting operator with n groups.
$\Pi(t)$	Time-dependent hierarchical policy, outputs (s_t, n_t).
s_t	Spatial unshuffle factor at timestep t.
n_t	Number of channel groups for codebook splitting at timestep t.
$\mathbf{K}_t$	Index maps at timestep t.
$\mathbf{K}_{t,i}$	ith channel of $\mathbf{K}_t$ at timestep t.
$\mathbf{c}_t^{\text{map}}$	Corrective map constructed from codebook at timestep t.

I. INTRODUCTION

REMOTE sensing imagery has become mission-critical across environmental monitoring, disaster response, and defense applications [1], [2], demanding efficient compression solutions due to the exponential growth in data volume. However, the deployment of advanced compression algorithms on satellite platforms faces unique constraints beyond simple rate-distortion (R-D) performance: adaptability, stability, and security.

Recent traditional image compression standards, such as BPG [3] and VVC [4], along with learned image compression (LIC) methods, including autoencoder-based approaches [5], [6], have significantly advanced the storage and transmission of visual data. However, these methods exhibit limited

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

perceptual quality at low bitrates, often producing blurry reconstructions that lack the fine textures and details critical for remote sensing (RS) interpretation tasks. To achieve higher compression ratios, codebooks—typically based on VQ-VAE architectures [7], [8]—are employed to construct shared knowledge bases. However, learned codebooks are inherently tied to the training data distribution, which introduces two critical limitations for RS applications: 1) learned codebooks are prone to collapse and struggle to generalize across the diverse and dynamic terrains of RS data (e.g., from urban areas to rainforests), necessitating frequent and costly retraining and 2) updating these models requires transmitting large weight parameters to orbit, which is impractical given the bandwidth constraints of satellite downlinks.

We propose spatio-temporal hierarchical codebook diffusion (STHCD), a novel framework for RS image compression (RSIC) that fundamentally rethinks resource allocation in the diffusion generation process. While our primary objective is compression (minimizing bitrate while maintaining perceptual fidelity), we implement this through a specific "coding" mechanism. Here, coding refers to the quantization of the diffusion process, where the continuous noise space is mapped to a discrete set of codebook indices for efficient transmission. Our approach is motivated by a critical observation: the diffusion denoising trajectory exhibits an inherent information hierarchy. Early stages (large t) predominantly recover macroscopic low-frequency structure, while later stages (small t) focus on reconstructing microscopic high-frequency texture. The significance and coding complexity of these two modalities differ drastically.

Inspired by Huffman coding principles where resource allocation is inversely proportional to redundancy, we recognize that this "coarse-to-fine" (CTF) hierarchy demands differentiated treatment. Early steps (low-frequency structure) are redundant and simple, while later steps (high-frequency texture) are complex. Yet existing codebook-based compression methods adopt a static and uniform brute-force strategy: whether in feature-space operations or in codebook design and matching, all spatial locations and all timesteps throughout the generation process are treated identically. This *one-size-fits-all* approach completely disregards the intrinsic hierarchy of information, leading to inefficient resource utilization.

STHCD addresses this limitation through two key innovations.

1) *Temporal Hierarchy:* We employ nonuniform diffusion timestep allocation, dedicating fewer computational resources to redundant early stages and concentrating capacity on complex later stages where fine details emerge.
2) *Spatial Hierarchy:* We introduce adaptive spatial allocation mechanisms that recognize varying information density across image regions, enabling efficient representation of both homogeneous and heterogeneous areas.

Our framework builds upon presampled, fixed codebooks derived from standard Gaussian distributions rather than learned feature codebooks. This design choice offers several structural advantages.

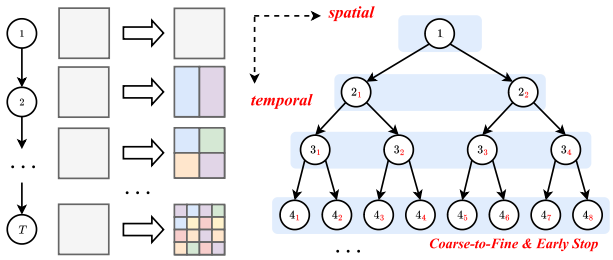


Fig. 1. Overview of the STHCD framework. Inspired by Huffman coding principles, STHCD maps temporal diffusion steps to hierarchical tree depths, enabling nonuniform rate allocation across space and time.

1) *Stability and Generalization:* Since the codebook represents generic Gaussian noise rather than specific image features, it is immune to codebook collapse and inherently generalizes to any scene type without retraining.
2) *Security and Decoupling:* The transmitted bitstream consists solely of indices pointing to random noise vectors. Unlike VQ-VAE tokens which carry semantic approximations, these indices are data-agnostic and statistically independent of the source image, offering an intrinsic layer of data obfuscation and security during transmission.

Similar concepts of temporal codebook structures have been explored in recent work such as DDCM [9], which is a temporal codebook diffusion (TCD) method. However, these methods employ naive, uniform time diffusion strategies with large timesteps, resulting in slow encoding efficiency and high storage overhead unsuitable for RS applications. In contrast, STHCD introduces spatio-temporal hierarchy to achieve both efficiency and quality.

As shown in Fig. 1, STHCD transforms the temporal diffusion process into a dynamic, hierarchical system. Crucially, our method is training-free. Instead of optimizing network weights, we implement rate control through logic-based command updates that adjust codebook granularity and residual shuffling on the fly. This "Plug-and-Play" nature allows STHCD to be deployed instantly on pretrained diffusion models, adapting to varying bandwidth constraints without the need for complex on-board retraining.

Diffusion models inherently reconstruct content in a CTF manner: early timesteps establish global low-frequency structures, while later timesteps refine local high-frequency details. However, existing TCD frameworks treat all timesteps equivalently, failing to exploit this natural hierarchy.

Our key insight is that *different frequency components require different encoding resources*. Low-frequency structures, being simpler and more predictable, can be encoded with fewer steps and coarser codebooks. In contrast, high-frequency details demand finer-grained codebooks for accurate reconstruction. This principle mirrors Huffman coding, where frequent symbols use shorter codewords and rare symbols use longer ones.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

The proposed STHCD framework transforms codebook diffusion from a static, uniform process into a dynamic, hierarchical system that intelligently adjusts its encoding strategy across different stages and features.

Guided by this philosophy, we design a novel complete codebook diffusion improvement framework requiring no additional training and seamlessly integrating with pretrained diffusion models. Our main contributions can be summarized as follows.

1) We propose a systematic hierarchical encoding strategy that dynamically adjusts encoding granularity across diffusion timesteps to align with the CTF reconstruction nature. Early timesteps use coarser representations with fewer steps for low-frequency structures, while later timesteps progressively refine with finer-grained representations for high-frequency details.
2) Residual unshuffle operations are introduced to decompose latent residuals into multigranularity components. This enables adaptive spatial resolution adjustments across different timesteps, enhancing the flexibility of the framework.
3) To achieve flexible bit allocation, a channel-wise codebook splitting strategy is proposed. By splitting codebooks along the channel dimension, this method avoids exponential memory growth while allowing fine-grained rate control.
4) We design a temporal orchestration mechanism that coordinates residual transformation and codebook splitting across diffusion timesteps, implementing nonuniform resource allocation tailored to content complexity at each stage.

In summary, by introducing hierarchical principles into codebook diffusion through time-dependent spatial transformations, we propose a systematic training-free framework that significantly enhances encoding efficiency for multiscale information in RS images, achieving superior compression performance with drastically reduced computational overhead.

The remainder of this article is organized as follows. Section II reviews related work. Section III details our proposed hierarchical codebook diffusion framework. Section IV presents comprehensive experimental results and analysis. Section V discusses STHCD's applicability across different data modalities and analyzes the limitations arising from using an RGB-pretrained diffusion backbone. Finally, Section VI concludes the article.

II. Related Work

Image compression has evolved from traditional codecs to learned neural approaches, with recent advances in generative models promising perceptual quality improvements at low bitrates. However, achieving efficient compression for complex, multiscale imagery—such as RS data—remains challenging due to the lack of adaptive resource allocation mechanisms. This section reviews three key research directions: learned and generative image compression methods, codebook-based discrete representations, and hierarchical coding strategies, examining their strengths and limitations in the context of our work.

A. Autoencoder-Based LIC

LIC has evolved from traditional autoencoder-based methods to generative models, aiming to achieve high perceptual quality at low bitrates. Early autoencoder architectures [5] pioneered end-to-end optimization with R-D objectives, achieving competitive performance at moderate bitrates. However, these methods struggled with perceptual quality at ultralow bitrates, producing blurry reconstructions that lack fine textures. Subsequent works [6] improved upon hyperprior models to enhance R-D tradeoffs, yet still faced challenges in preserving textures at extremely low bitrates. To address these limitations, generative adversarial networks (GANs) [10], [11] were introduced to synthesize realistic textures. However, GANs suffered from training instability and mode collapse, limiting their reliability for compression tasks.

Recent research in RSIC has primarily focused on adapting network architectures to the specific content of aerial and satellite imagery. These efforts include leveraging edge information [12], exploring spatial–temporal context [13], and maintaining object-level fidelity [14]. Several works have begun to integrate diffusion models to enhance perceptual quality. For instance, some approaches use diffusion as a postprocessing enhancer [15], [16], which, while effective, often fails to integrate the generative process into a unified, flexible encoding framework. This typically results in systems requiring multiple models for different bitrates, increasing storage overhead.

Beyond visible-light RS imagery, recent hyperspectral image (HSI) studies have further highlighted the importance of jointly modeling spatial structure, spectral consistency, and scene transferability. Representative advances include interactive spatial-context learning for HSI classification [17], semantic constraint-based spatial–spectral multiview clustering [18], and cross-scene self-training strategies such as DHSNet [19], all of which demonstrate that exploiting modality-specific structure is critical for robust HSI processing.

B. Diffusion-Based Generative Image Compression

Diffusion models [20], [21], [22] have been adapted for image compression through two primary approaches: continuous conditional generation and discrete codebook-based methods. Continuous conditional diffusion [23] encodes images into compact latent representations to guide the diffusion process, achieving high perceptual quality. However, this approach places the entire synthesis burden on the decoder, resulting in excessive computational complexity unsuitable for resource-constrained environments. The TCD paradigm [24] addresses this by replacing random noise with discrete codebook selections, forming a compressed bitstream of indices.

Discrete representation is central to modern generative compression. The dominant approach relies on vector quantized, which learn a codebook by clustering feature vectors from a training dataset. While effective for specific domains, these methods suffer from codebook collapse (where indices are underutilized) and domain overfitting. In RS scenarios, codebook and model trained on urban data may fail to reconstruct forests efficiently, limiting its generalization. In TCD, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF RELATED WORK IN LIC

Domain	Year	Ref	Contributions
Autoencoder-based	2023	[29]	Introduced progressive context-based trit-plane coding for quality scalability.
	2023	[12]	Preserved edges/textures with an edge-guided adversarial network (EGA-Net) and edge fidelity constraint.
	2024	[14]	Proposed an OF-RSIC method for low-bitrate object-fidelity remote sensing image compression, integrating region splitting, dual attention modules, and region-differentiated loss.
Diffusion-based	2023	[23]	Introduced conditional diffusion models for lossy image compression.
	2024	[16]	Applied diffusion models to RS image compression as post-processing.
	2025	[24]	Introduced denoising diffusion codebook models into RS image compression.
	2025	[15]	Applied diffusion prior to enhance reconstructed RS imagery.
Our Method	Distinct Contributions and Strengths		
	Time-dependent hierarchical policy: Allocates rate across diffusion stages, optimizing compression performance without retraining. **Spatial and channel-wise transformations:** Combines spatial residual transformation and codebook splitting to dynamically capture spatial variations, reducing redundancy and improving compression efficiency. **Progressive discrete coding via the diffusion process:** Leverages the diffusion process itself for coding, inherently aligning with its progressive nature, enabling efficient and scalable compression.		

codebook is a fixed set of Gaussian noise vectors. This approach decouples the codebook from the data distribution, guaranteeing: 1) no training instability or collapse; 2) universal generalization across datasets; and 3) content-agnostic transmission, where the bitstream reveals nothing about the image semantics without the specific decoder key (seed/schedule), enhancing security. However, existing TCD implementations employ a uniform codebook strategy, leading to inefficiencies.

C. Hierarchical Image Compression

Hierarchical and progressive coding have long been recognized as essential for efficient compression, enabling quality scalability and adaptive bitrate allocation. Traditional codecs such as JPEG [25] and JPEG2000 [26] implement hierarchical structures to prioritize important information. Progressive neural compression has predominantly been achieved through the paradigm of "non-linear transform coding + progressive quantization." Lu et al. [27] achieved quality scalability within a single bitstream via nested quantization. Lee et al. [28] and Jeon et al. [29] adopted trit-plane coding, enabling fine-grained scalability in bitrate adjustment while achieving R-D performance comparable to single-rate methods. Meanwhile, the unified framework proposed by Li et al. [30] and [31] took a more holistic optimization approach, flexibly balancing bitrate, distortion, and computational complexity within a single model to accommodate diverse deployment needs.

However, current diffusion-based compression frameworks, especially TCD, produce flat bitstreams that lack hierarchical structure. This is particularly problematic for RS imagery, which inherently contains multiscale information with vast homogeneous regions and complex, detail-rich areas. A uniform coding strategy inevitably leads to wasteful bit allocation, either overallocating bits to simple regions or underallocating them to critical structures. While recent RSIC work [15], [16] applied diffusion models for enhancement, they fail to address the fundamental inefficiency of uniform encoding in the diffusion process itself. This exposes a critical gap: *the lack of a hierarchical framework that aligns the diffusion denoising process with principles of adaptive information coding*.

This article addresses this gap by proposing STHCD, a training-free framework inspired by Huffman coding principles. By introducing spatial residual transformation, channel-wise codebook splitting, and time-dependent hierarchical policies, STHCD dynamically allocates coding resources across diffusion stages, achieving state-of-the-art compression performance on multiscale RS imagery without retraining. Table I provides a comprehensive comparison of related work, highlighting the distinct contributions of our proposed method.

III. PROPOSED METHOD

A. Motivation and Overview

Contemporary TCD employs a uniform encoding strategy: at every diffusion timestep, a fixed spatial processing granularity and a single discrete codebook are applied uniformly across the entire image representation. This homogeneous treatment introduces two critical inefficiencies: 1) *coding redundancy:* as the model allocates identical computational resources to both simple low-frequency structures and complex high-frequency details and 2) *prohibitive latency:* with decoding time scaling linearly with a large number of steps (e.g., $T = 500 \sim 1000$).

The root cause lies in a fundamental mismatch between the diffusion process and the encoding strategy. Diffusion models inherently reconstruct content in a CTF manner: early timesteps rapidly establish global low-frequency structures (e.g., overall layout, dominant colors), while later timesteps progressively refine local high-frequency details (e.g., textures, edges). However, TCD treats all timesteps equivalently—analogous to using the same brush size for both painting large background areas and fine details. Fig. 1 illustrates this mismatch: uniform resource allocation fails to exploit the natural information hierarchy along the temporal axis.

Our key observation is straightforward: *different frequency components require different amounts of encoding resources*. Early steps are dominated by low-frequency structures with high spatial redundancy. Low-frequency structures, being relatively simple and predictable, can be encoded with fewer steps and coarser codebooks. Conversely, high-frequency details, exhibiting higher complexity and variability, demand more

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

steps and finer-grained codebooks for accurate reconstruction. The key symbols and their descriptions used throughout this article are summarized in Nomenclature.

To implement this nonuniform allocation, we introduce the STHCD framework, comprising three controllable components.

1) *Spatial Residual Transformation:* Decomposes the residual representation into multigranularity components via operations. For example, a $2\times$ residual unshuffle converts a $\mathbf{r}_t \in \mathbb{R}^{H\times W\times C}$ residual map into $\mathbf{r}'_t \in \mathbb{R}^{H/2\times W/2\times 4C}$, effectively increasing spatial resolution at the cost of channel expansion. Critically, this operation is zero-parameter, incurring no additional memory overhead for the transformation itself. The model adaptively adjusts spatial granularity: coarse-grained (no unshuffle) for early low-frequency phases, fine-grained (with unshuffle) for later high-frequency phases.
2) *Spatial Codebook Splitting:* Splits the codebook along the channel dimension to enable variable bit allocation per step. For instance, a unified 4-channel codebook can be split into four 1-channel codebooks, allowing independent quantization of each channel. This splitting increases the number of codebook lookups per step (from 1 to 4) but reduces the combinatorial space per lookup (from $Q_4 = |\mathbf{c}|^4$ effective combinations to $Q_1 = |\mathbf{c}|^1$), providing finer control over R-D tradeoffs. The split codebooks share the same embedding matrix, so memory grows only linearly in the split factor, not exponentially. Early phases use fewer splits (coarse, e.g., 4-ch as a single block), while later phases use more splits (fine, e.g., 1-ch per group).
3) *Time-Dependent Hierarchical Policy:* Orchestrates the temporal scheduling of residual transformation and codebook splitting. The policy $\Pi(t)$ specifies, for each timestep t, the configuration of spatial granularity and codebook grouping.

STHCD generalizes TCD as a special case. When residual transform and codebook splitting are set to identity transformations (no spatial decomposition, no codebook splitting) and $\Pi(t)$ is uniform across all timesteps, STHCD reduces exactly to the standard TCD formulation. Formally, the expressive capacity of TCD forms a proper subset of STHCD's representation space: $\mathcal{F}_{\text{TCD}} \subset \mathcal{F}_{\text{STHCD}}$.

Empirically, under a fixed index budget (total number of codebook lookups), STHCD achieves superior reconstruction quality with significantly fewer diffusion steps. For instance, where TCD requires 1000 or 800 steps, STHCD achieves comparable or better quality with only 125, 80, or even 15 steps, respectively, directly reducing decoding latency and computational overhead.

1) Clarification on Index Budget: Throughout this work, we maintain a constant *index budget* rather than a strict physical bitrate budget. Specifically, we fix the total number of codebook lookups (indices) across the entire diffusion trajectory. For fair comparison, we fix the codebook size $K = 256$ across all experiments, ensuring that the total bits is proportional to the number of indices. Thus, "constant index budget" implies constant bitrate when K is fixed. The total physical bitrate in bits can be computed as follows:

$$\text{Total bits} = \sum_i \left(n_i \cdot \log_2 K_i\right) \tag{1}$$

where n_i is the number of indices in phase i and $|K_i|$ is the codebook size for that phase. If the codebook size varies across phases (due to codebook splitting), the physical bitrate will also vary accordingly. For applications requiring strict bit-level budget control, one can fix the codebook size to a power-of-two (e.g., $|K| = 2^k$) and adjust only the number of lookups, ensuring constant bit count. Our experiments adopt the former (fixed index count) for simplicity, but the framework naturally accommodates both constraints.

B. TCD for Image Representation

The compression process of TCD begins by mapping the high-dimensional input image $\mathbf{x}_0$ into a compact and information-rich latent representation, $\mathbf{z}_0$, using a pretrained variational autoencoder (VAE) encoder $\mathcal{E}$. All subsequent compression operations are performed within this latent space to ensure computational efficiency. The corresponding VAE decoder $\mathcal{D}$ reconstructs the image from the final latent variable

$$\mathbf{z}_0 = \mathcal{E}\left(\mathbf{x}_0\right), \quad \hat{\mathbf{x}}_0 = \mathcal{D}\left(\hat{\mathbf{z}}_0\right). \tag{2}$$

Diffusion models have demonstrated remarkable success in generating high-quality images by iteratively denoising a latent variable. At each timestep t, the reverse process is defined as follows:

$$\hat{\mathbf{z}}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\hat{\mathbf{z}}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta\left(\hat{\mathbf{z}}_t, t\right)\right) + \sigma_t \boldsymbol{\varepsilon} \tag{3}$$

where ϵ_θ predicts the noise, and $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \mathbf{I})$ introduces stochasticity for diverse generation. This process enables the model to synthesize realistic images by progressively refining noisy latent variables.

In the context of image compression, the objective shifts from generating diverse outputs to reconstructing a specific latent representation $\mathbf{z}_0$ with high fidelity. To achieve this, the stochastic noise term $\sigma_t \mathbf{z}$ is replaced with a deterministic selection from a discrete codebook $\mathbf{c}_{t-1}$, which is presampled from $\mathcal{N}(0, \mathbf{I})$. The reverse update thus becomes

$$\hat{\mathbf{z}}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\hat{\mathbf{z}}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta\left(\hat{\mathbf{z}}_t, t\right)\right) + \sigma_t \mathbf{c}_{t-1}^{(k_t)} \tag{4}$$

$$= \mu_t\left(\hat{\mathbf{z}}_t\right) + \sigma_t \mathbf{c}_{t-1}^{(k_t)} \tag{5}$$

where $\mathbf{c}_{t-1}^{(k_t)}$ is the codebook vector selected to best approximate $\mathbf{z}_0$. The sequence of indices $\{k_t\}$ is recorded, converting the continuous denoising process into a discrete and compressible bitstream.

At each timestep, the goal of TCD is to find the best codebook index k_t that minimizes the residual between the predicted clean latent $\hat{\mathbf{z}}_{0|t}$ and the actual latent $\mathbf{z}_0$. This is achieved by

$$k_t = \arg\min_k \mathcal{L}_{\text{TCD}}\left(\mathbf{z}_0, \hat{\mathbf{z}}_{0|t}, \mathbf{c}_{t-1}, k\right) \tag{6}$$

$$= \arg\min_k \left\| \mathbf{c}_{t-1}^{(k)} - \sigma_t \frac{\sqrt{\bar{\alpha}_t}}{1-\bar{\alpha}_t}\left(\mathbf{z}_0 - \hat{\mathbf{z}}_{0|t}\right) \right\|^2 \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

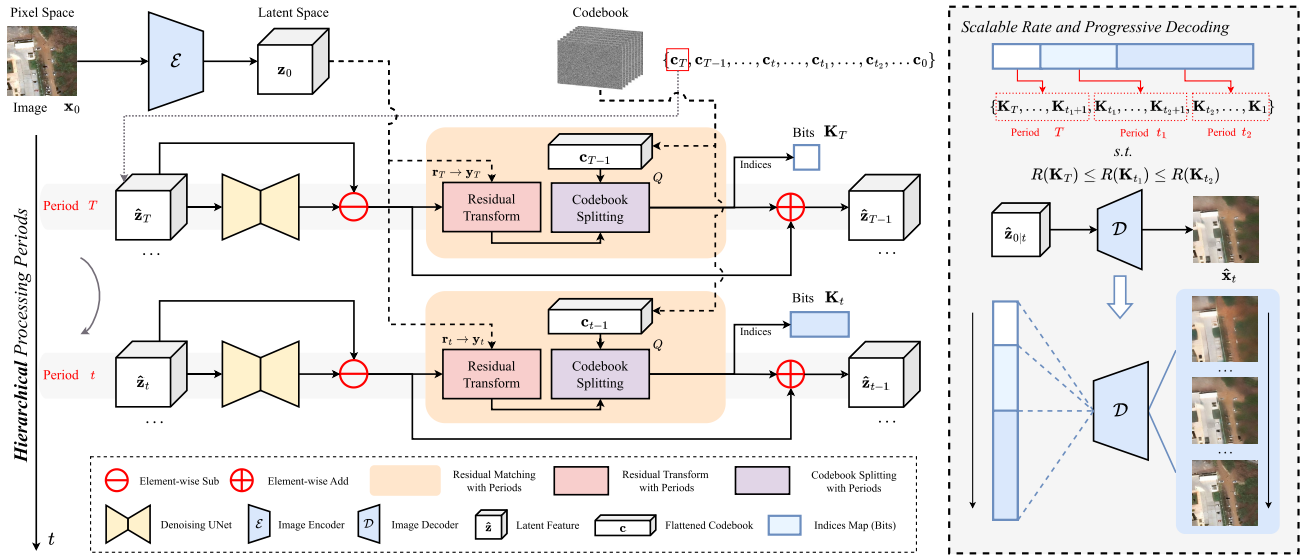


Fig. 2. Schematic illustration of the proposed STHCD framework. The pipeline consists of three phases: 1) *latent initialization:* where the input image $\mathbf{x}_0$ is encoded into a compact latent representation $\mathbf{z}_0$; 2) *hierarchical residual encoding:* which iteratively processes the latent features across diffusion periods $T, \ldots, t$. In each period, the Spatial Residual Transformation and Codebook Splitting modules dynamically decompose residuals into multigranularity components and allocate bits $\mathbf{K}_t$ based on the content complexity; and 3) *scalable decoding:* shown on the right, where the decoder $\mathcal{D}$ reconstructs the image $\hat{\mathbf{x}}_t$ progressively, achieving scalable rate control $R(\mathbf{K}_t)$ and CTF quality refinement.

$$= \arg\max_k \left\langle \mathbf{c}_{t-1}^{(k)}, \mathbf{z}_0 - \hat{\mathbf{z}}_{0|t} \right\rangle. \tag{8}$$

C. Spatio-Temporal Hierarchical Codebook Diffusion

As illustrated in Fig. 2, the proposed STHCD framework establishes a CTF generative compression pipeline. The overall procedure is orchestrated by a Time-Dependent Hierarchical Policy, which dynamically allocates computational and bit-rate resources across different diffusion timesteps. The workflow proceeds as follows. First, the input image $\mathbf{x}_0$ is mapped to a latent space via the encoder $\mathcal{E}$ to obtain $\mathbf{z}_0$. The encoding process is then discretized into hierarchical processing periods (e.g., Period T, Period t_1, Period t_2), aligning with the diffusion model's nature of progressively resolving structural details.

Within each processing period, a Denoising UNet predicts the current feature estimate $\hat{\mathbf{z}}_t$. The system calculates the difference between the current latent and the prediction to obtain the residual. To efficiently compress this information, the residual passes through the Residual Matching module, which comprises two key operations. First, the residual is decomposed into multigranularity components via unshuffle operations, adapting the spatial resolution to the current timestep's frequency characteristics. Second, the transformed residual is quantized using a channel-wise codebook splitting strategy. This generates the bitstream indices (Bits $\mathbf{K}_t$) while preventing exponential memory growth. The quantized residual is then added back to update the latent state for the subsequent period ($\hat{\mathbf{z}}_{t-1}$), ensuring strictly consistent feature evolution.

As depicted in the right panel of Fig. 2, the bitstream supports scalable decoding. The decoder $\mathcal{D}$ can reconstruct the image $\hat{\mathbf{x}}_t$ at any intermediate stage. Earlier periods (e.g., Period T) provide low-frequency structural information with fewer bits, while later periods (e.g., Period t_2) progressively refine high-frequency details, satisfying the rate constraint $R(\mathbf{K}_T) \le R(\mathbf{K}_{t_1}) \le R(\mathbf{K}_{t_2})$.

1) Temporal Staging Strategy: The foundation of STHCD is the partitioning of the total diffusion timesteps, T, into a series of sequential stages. Each stage corresponds to a specific interval of timesteps and is assigned a distinct set of spatial encoding parameters. This temporal segmentation forms the backbone of our hierarchical scheme, enabling the subsequent spatial strategies to vary dynamically as the latent state $\mathbf{z}_t$ transitions from pure noise toward a clean representation. The parameters for each stage are determined by a predefined schedule, $\Pi(t)$, which maps a given timestep t to a specific configuration. To achieve spatio-temporal synergy, we introduce hierarchical partitioning in the spatial domain, which operates at both the residual level and the codebook level. The selection of a specific spatial strategy is dictated by the current temporal stage.

2) Residual Spatial Transformation: To regulate the information density across different diffusion stages, we employ a parameter-free residual unshuffle operation, this operation reshapes the spatial structure of the residual map $\mathbf{r}_t \in \mathbb{R}^{H\times W\times C}$ by trading spatial resolution for channel depth. Specifically, for a downsampling factor s, the transformation is defined as follows:

$$\mathbf{r}'_t = \downarrow_{s_t} (\mathbf{r}_t), \quad \text{where } \mathbf{r}'_t \in \mathbb{R}^{\frac{H}{s}\times\frac{W}{s}\times(s^2\cdot C)}. \tag{9}$$

Visually, this process rearranges spatially neighboring $s \times s$ blocks into the channel dimension, effectively preserving all information while adapting the feature granularity. The inverse operation, *Residual Shuffle* ($\uparrow_{s_t} = \downarrow_{s_t}^{-1}$), losslessly restores the original dimensions.

It is important to emphasize that the spatial unshuffle operation is strictly confined to the residual quantization branch. The diffusion UNet always takes the latent $\hat{\mathbf{z}}_t$ in its original spatial resolution as input [as shown in (5)]. The quantized

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

residual is immediately reshuffled back to the original layout ($\uparrow_{s_t}$) before updating the latent state. Consequently, the UNet consistently operates on spatially continuous features, ensuring that its convolutional inductive biases regarding local neighborhood structures are preserved and unaffected by the internal granularity adjustments of the codebook module.

3) Codebook Spatial Splitting: To achieve flexible bit allocation, we introduce a dynamic Codebook Splitting strategy driven by the diffusion timestep t. Instead of quantizing the entire C-dimensional vector holistically, we partition the channel dimension into $\max(n_t)$ subgroups based on the current content complexity. The quantization operator Q_{n_t} slices the residual $\mathbf{r}_t$ along the channel axis

$$Q_{n_t}(\mathbf{r}_t) = \left[Q\left(\mathbf{r}_{t,1}\right), Q\left(\mathbf{r}_{t,2}\right), \ldots, Q\left(\mathbf{r}_{t,n_t}\right)\right] \tag{10}$$

where $\mathbf{r}_{t,i}$ represents the ith subvector of dimension C/n_t. Each sub-vector is independently quantized by finding the nearest neighbor in the shared codebook $\mathbf{c}_{t-1}$

$$\mathbf{K}_{t,i} = \arg\max_{\mathbf{e}\in\mathbf{c}_{t-1}} \langle \mathbf{r}_{t,i}, \mathbf{e}\rangle. \tag{11}$$

Here, $\mathbf{K}_{t,i}$ denotes the codebook index for the ith split. By adjusting n_t across timesteps, the system dynamically controls the bit-rate $R(\mathbf{K}_t) \propto n_t$, allocating more bits to stages requiring fine-grained details.

Algorithm 1 STHCD Encoding Process

Input: Target image $\mathbf{x}_0$, Encoder $\mathcal{E}$, Noise predictor $\epsilon_\theta(\cdot, t)$, Timesteps T, Codebooks $\mathbf{c}$, Spatio-temporal schedule $\Pi(t)$.
Output: Index maps $[\mathbf{K}_T, \mathbf{K}_{T-1}, \ldots, \mathbf{K}_1]$.

$\mathbf{z}_0 \leftarrow \mathcal{E}(\mathbf{x}_0)$;
Initialize $\hat{\mathbf{z}}_T$ with $\mathbf{c}_T$;
for $t = T, T-1, \ldots, 1$ **do**
  $(s_t, n_t) \leftarrow \Pi(t)$;
  $\boldsymbol{\epsilon}_\theta^{\text{pred}} \leftarrow \epsilon_\theta(\hat{\mathbf{z}}_t, t)$;
  $\hat{\mathbf{z}}_{0|t} \leftarrow \frac{1}{\sqrt{\bar{\alpha}_t}}(\hat{\mathbf{z}}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}_\theta^{\text{pred}})$;
  $\mathbf{r}_t \leftarrow \mathbf{z}_0 - \hat{\mathbf{z}}_{0|t}$;
  // Apply hierarchical spatial encoding
  $\mathbf{y}_t \leftarrow \downarrow_{s_t}(\mathbf{r}_t)$ // Eq. (12)
  $\mathbf{K}_t \leftarrow Q_{n_t}(\mathbf{y}_t)$ // Eqs. (13), (10), (11)
  $\hat{\mathbf{y}}_t \leftarrow \mathbf{c}_{t-1}(\mathbf{K}_t)$;
  $\mathbf{c}_{t-1}^{\text{map}} \leftarrow \uparrow_{s_t}(\hat{\mathbf{y}}_t)$ // Eq. (14)
  // Update latent state (spatially restored)
  $\hat{\mathbf{z}}_{t-1} \leftarrow \frac{1}{\sqrt{\alpha_t}}(\hat{\mathbf{z}}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta^{\text{pred}}) + \sigma_t \mathbf{c}_{t-1}^{\text{map}}$;
end
return $[\mathbf{K}_T, \mathbf{K}_{T-1}, \ldots, \mathbf{K}_1]$;

4) STHCD Encoding and Decoding: The complete STHCD encoding process is detailed in Algorithm 1. The core operation within each temporal stage involves transformations defined by the schedule $\Pi(t) \to (s_t, n_t)$. Given the residual between the ground truth (GT) latent $\mathbf{z}_0$ and the current estimate $\hat{\mathbf{z}}_{0|t}$, we first apply the spatial transformation

$$\mathbf{y}_t = \downarrow_{s_t}\left(\mathbf{z}_0 - \hat{\mathbf{z}}_{0|t}\right). \tag{12}$$

Next, the transformed residual $\mathbf{y}_t$ is quantized using the channel-wise operator to produce a set of index maps $\mathbf{K}_t$

$$\mathbf{K}_t = Q_{n_t}(\mathbf{y}_t). \tag{13}$$

These indices are used to retrieve the corresponding codebook vectors, forming a quantized representation $\hat{\mathbf{y}}_t = \mathbf{c}_{t-1}(\mathbf{K}_t)$. Finally, this representation is transformed back to the original latent space dimensionality to create the corrective map $\mathbf{c}_{t-1}^{\text{map}}$

$$\mathbf{c}_{t-1}^{\text{map}} = \uparrow_{s_t}(\hat{\mathbf{y}}_t). \tag{14}$$

Algorithm 2 STHCD Decoding Process

Input: Index maps $[\mathbf{K}_T, \ldots, \mathbf{K}_1]$, Decoder $\mathcal{D}$, Noise predictor $\epsilon_\theta(\cdot, t)$, Codebooks $\mathbf{c}$, Spatio-temporal schedule $\Pi(t)$.
Output: Reconstructed image set $\{\hat{\mathbf{x}}_{T-1}, \ldots, \hat{\mathbf{x}}_0\}$.

Initialize latent $\hat{\mathbf{z}}_T$ with $\mathbf{c}_T$;
for $t = T, T-1, \ldots, 1$ **do**
  $(s_t, n_t) \leftarrow \Pi(t)$;
  Retrieve index map $\mathbf{K}_t$;
  // Reconstruct corrective map from indices
  $\hat{\mathbf{y}}_t \leftarrow \mathbf{c}_{t-1}(\mathbf{K}_t)$;
  $\mathbf{c}_{t-1}^{\text{map}} \leftarrow \uparrow_{s_t}(\hat{\mathbf{y}}_t)$;
  // Perform guided reverse diffusion step
  $\boldsymbol{\epsilon}_\theta^{\text{pred}} \leftarrow \epsilon_\theta(\hat{\mathbf{z}}_t, t)$;
  $\hat{\mathbf{z}}_{t-1} \leftarrow \frac{1}{\sqrt{\alpha_t}}(\hat{\mathbf{z}}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta^{\text{pred}}) + \sigma_t \mathbf{c}_{t-1}^{\text{map}}$;
  // Progressive decoding to image space
  if $t = T-1, T-2, \ldots, 1$ **then**
    $\hat{\mathbf{x}}_t \leftarrow \mathcal{D}(\hat{\mathbf{z}}_{0|t})$;
  end
end
$\hat{\mathbf{x}}_0 \leftarrow \mathcal{D}(\mathbf{z}_0)$;
return $\{\hat{\mathbf{x}}_{T-1}, \ldots, \hat{\mathbf{x}}_0\}$;

The STHCD decoding process, shown in Algorithm 2, deterministically reconstructs the image from the bitstream of index maps $[\mathbf{K}_T, \ldots, \mathbf{K}_1]$. Starting from the same fixed noise latent $\mathbf{z}_T$, the decoder iterates backward in time. At each step t, it retrieves the corresponding index map $\mathbf{K}_t$, looks up the codebook vectors, and applies the scheduled inverse spatial transforms ($\uparrow_{s_t}$) to construct the corrective map $\mathbf{c}_{t-1}^{\text{map}}$. This map is then used in the reverse diffusion update to compute $\mathbf{z}_{t-1}$. This guided process is repeated until the final latent $\mathbf{z}_0$ is obtained, which is then passed to the decoder $\mathcal{D}$ to yield the reconstructed image $\hat{\mathbf{x}}_0$.

5) Relation to TCD: The STHCD framework serves as a generalization of the conventional TCD approach. TCD can be viewed as a special instance of STHCD where the spatio-temporal schedule $\Pi(t)$ is static and uniform across all timesteps. Specifically, TCD is equivalent to setting the hierarchical parameters to their most basic configuration for all t, with no feature-level spatial transform ($s_t \equiv 1$) and no codebook-level channel partitioning ($n_t \equiv 1$). Under these constraints, the quantization in STHCD simplifies to

$$\hat{\mathbf{x}}_t^{\text{STHCD}} \to \mathcal{D}\left(\mathbf{c}_{t-1}\left(Q\left(\mathcal{E}\left(\mathbf{x}_t\right)\right)\right)\right) \equiv \hat{\mathbf{x}}_t^{\text{TCD}} \tag{15}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

demonstrating that TCD is a specific, nonhierarchical case within our more flexible framework. Fig. 3 illustrates the TCD single-step denoising and codebook matching process for this special case.

For scalable coding, we use the following formulations. The complete decoding bits per pixel (bpp) for our method is computed as follows:

$$R_{\mathrm{TCD}}(t) = \frac{(T-t) \times \log_2 K}{H \times W} \tag{16}$$

where T is the total timesteps, K is the codebook size, $H \times W$ represents the image dimensions, and t denotes the current decoding step for progressive encoding. For TCD scheme, we use $T = 1000, K = 256$ for rate and distortion balance.

For the proposed STHCD scheme, we calculate the total bpp by accounting for the spatio-temporal hierarchical encoding at each timestep. At timestep t, the spatial unshuffle operation with factor s_t transforms the residual map from $(H, W, C_{\mathrm{base}})$ to $(H/s_t, W/s_t, C_{\mathrm{base}} \cdot s_t^2)$, and the codebook splitting operator Q_{n_t} quantizes n_t channel groups independently. The total bpp is given by

$$R_{\mathrm{STHCD}} = \frac{1}{H \times W} \sum_{t=1}^{T} \left(\frac{n_t}{s_t^2} \times \log_2 K \right) \tag{17}$$

where s_t and n_t are determined by the time-dependent hierarchical policy $\Pi(t) \to (s_t, n_t)$, and $|K|$ is the codebook size. This can be simplified as follows:

$$R_{\mathrm{STHCD}} = \log_2 K \sum_{t=1}^{T} \frac{n_t}{s_t^2}. \tag{18}$$

IV. Experiments

A. Implementation Details

The proposed STHCD framework, which incorporates the Time-Dependent Hierarchical Policy, Spatial Residual Transformation, and Spatial Codebook Splitting, is implemented based on the Stable Diffusion 2.1 with diffusers.[1] During inference, we utilize DDPM sampling with varying numbers of denoising steps for encoding and decoding. All experiments are conducted on a single NVIDIA GeForce RTX 4090 GPU.

1) Datasets: We evaluate our method on a variety of RS imagery datasets, including RSICD [32], UCMerced [33], Real-RSSRD [34], AID [35], VRSBench [36], DOTA [37], [38], fMoW [39], DIOR [40], and NWPU-VHR-10 [41], [42]. 128 images with different classes are randomly selected and mixed from each dataset for test, with all images cropped and resized to a unified resolution of 256 × 256. These datasets cover a wide range of categories and scenarios, ensuring high diversity and representativeness for evaluation.

2) Evaluation Metrics: Multiple metrics are evaluated to comprehensively assess compression performance, including learned perceptual image patch similarity (LPIPS) [43], deep image structure and texture similarity (DISTS) [44], and Fréchet inception distance (FID) [45]. LPIPS and DISTS emphasize texture preservation and perceptual fidelity, while FID measures perceptual realism by comparing the feature distributions of original and reconstructed image sets. Unlike LPIPS and DISTS, which operate on individual image pairs, FID evaluates distribution-level similarity. Following [10], we calculate FID using patches extracted from each $H \times W$ image. Specifically, $\lfloor H/f \rfloor \cdot \lfloor W/f \rfloor$ nonoverlapping $f \times f$ patches are extracted, and additional $(\lfloor H/f \rfloor - 1) \cdot (\lfloor W/f \rfloor - 1)$ patches are obtained by shifting the extraction windows by $f/2$. These metrics collectively provide a comprehensive evaluation of perceptual realism and fidelity in image compression.

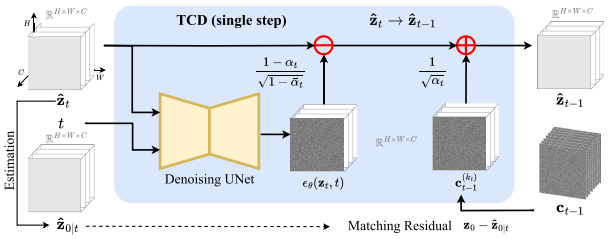


Fig. 3. Overviewof the TCD single-step denoising and codebook matching process. At each timestep, the model predicts the clean latent $\hat{\mathbf{z}}_{0|t}$ from the noisy latent $\hat{\mathbf{z}}_t$. The residual between $\hat{\mathbf{z}}_{0|t}$ and the target latent $\mathbf{z}_0$ guides the selection of the optimal codebook vector $\mathbf{c}_{t-1}^{(k_t)}$ to minimize reconstruction error. This selected codebook vector is then used in the reverse update to obtain $\hat{\mathbf{z}}_{t-1}$.

3) Baseline Methods: To evaluate the effectiveness of the proposed method, we compare it with both traditional codecs and state-of-the-art learning-based methods. The traditional codecs include JPEG [25], JPEG2000 [26], WebP [46], BPG,[2], and VTM-23.11,[3] where WebP is implemented using the Pillow (PIL) library. For learning-based methods, we include HiFiC [10] (GAN-based), ELIC [6] (VAE-based), CDC [23] (Diffusion-based), TCM [47] (Transformer-based) and CTC [29] (Progressive coding), with evaluations conducted using the CompressAI framework.[4] LDM-RSIC [15] as state-of-the-art RS compression method. Additionally, we implement the TCD method as a direct baseline to assess the improvements brought by our STHCD framework. All learning-based methods are retrained on the same RS image dataset to ensure a fair comparison.

4) Downstream Task: To demonstrate the utility of our perceptual compression method for downstream tasks, we conduct experiments on RS image object detection using official pretrained YOLOv11 [48] model.[5] The performance is evaluated using mean average precision (mAP) [49], [50] and $F1$ score [51] metrics. Higher mAP values indicate better detection performance, reflecting the ability of the compressed images to preserve critical features for object recognition, while higher $F1$ score indicates a better balance between precision and recall.

B. Main Results

As shown in Fig. 4, we compare the proposed STHCD method with the baseline TCD and several advanced image compression methods, including traditional approaches such as VVC, as well as learning-based methods like ELIC, HiFiC,

[1] https://github.com/huggingface/diffusers

[2] https://bellard.org/bpg

[3] https://vcgit.hhi.fraunhofer.de/jvet/VVCSoftware_VTM/-/tree/VTM-23.11

[4] https://github.com/InterDigitalInc/CompressAI

[5] https://docs.ultralytics.com/models/yolo11

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

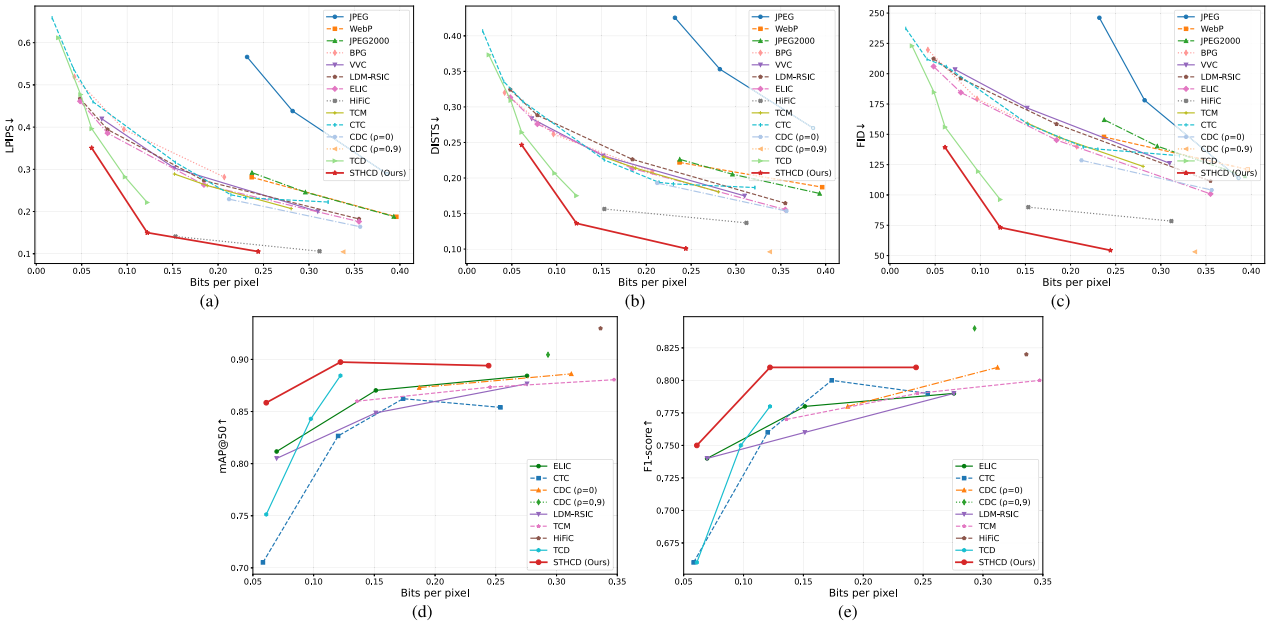


Fig. 4. R-D curves and downstream detection task performance on the RS visible-light dataset. (a)–(c) Perceptual quality metrics LPIPS, DISTS, and FID at different bitrates (bpp), where lower values indicate better perceptual quality of the reconstructed images. (d) and (e) mAP@50 and $F1$-score results for object detection tasks on the DOTA dataset using pretrained models, where higher values indicate better performance. STHCD (Ours) refers to the proposed method (red solid line), and TCD represents the baseline method. The results demonstrate that STHCD significantly outperforms the baseline and other mainstream compression methods in both perceptual quality and detection accuracy.

and CDC. The experimental results validate the effectiveness of STHCD from two perspectives: perceptual quality and downstream task performance.

In Fig. 4(a)–(c), we evaluate three perceptual quality metrics: LPIPS, DISTS, and FID. STHCD (red curve) demonstrates significant performance improvements, consistently appearing in the lower-left corner of all three plots. This indicates that STHCD achieves the lowest perceptual error at the same bitrate (bpp) or requires significantly fewer bits to achieve the same perceptual quality. Compared to the baseline (light green curve), STHCD exhibits substantial improvements, particularly in the low-bitrate range (0.05–0.15 bpp). In this range, the steeper decline in LPIPS and DISTS for STHCD highlights the effectiveness of the proposed modules in enhancing feature representation, enabling the reconstruction of textures and structures that align better with human perception.

Fig 4(d) and (e) presents the object detection performance (mAP@50 and $F1$-score) on the DOTA dataset. STHCD achieves a superior rate-accuracy tradeoff, with its curves positioned in the upper-left corner of the plots, significantly outperforming other methods. This demonstrates that STHCD preserves critical semantic information for object detection even at extremely low bitrates. Notably, at bitrates around 0.05 bpp, other methods such as CTC and TCD suffer severe degradation in mAP and $F1$-score, whereas STHCD maintains high detection accuracy (mAP $>$ 0.85). This result highlights that STHCD not only optimizes visual quality but also successfully retains key discriminative features in the images. Compared to the baseline, STHCD achieves a significant improvement in mAP at the same bitrate, demonstrating its superior capability in preserving complex object features in RS images.

C. Qualitative Results Analysis

To intuitively verify the effectiveness of the proposed method, Figs. 5 and 6 provide a visual reconstruction comparison between STHCD and current mainstream LIC methods (including CDC, CTC, ELIC, TCM, and LDM-RSIC) under different experimental settings.

Fig. 5 first illustrates the comparison under similar perceptual quality conditions (Iso-Quality). While the reconstructed images exhibit comparable visual fidelity across all methods (addressing the reviewer's observation of similarity), the quantitative advantages are substantial. As shown in the newly added bitrate ratio charts on the right, STHCD achieves these quality levels with significantly lower bandwidth consumption. For instance, in the dense building scenario (second row), our method achieves an optimal LPIPS of 0.0960 with only 0.1221 bpp, whereas baselines like CDC and CTC require +160% to +198% more bitrate to reach a similar visual standard. This confirms the superior coding efficiency of our approach.

To further distinguish the reconstruction capabilities, Fig. 6 presents a comparison under a strict low-bitrate constraint (Iso-Bitrate, bpp $\approx$ 0.15). In this challenging setting, the performance gap becomes visually evident. Typical baselines such as ELIC and TCM succumb to severe oversmoothing and detail loss, particularly in high-frequency regions like the riverbed textures in the first row. In contrast, STHCD maintains sharp boundaries and realistic fine-grained textures (e.g., the distinct roof tiles in the second row), consistent with its significantly lower (better) LPIPS and DISTS scores highlighted

[6]https://github.com/nachifur/MulimgViewer

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

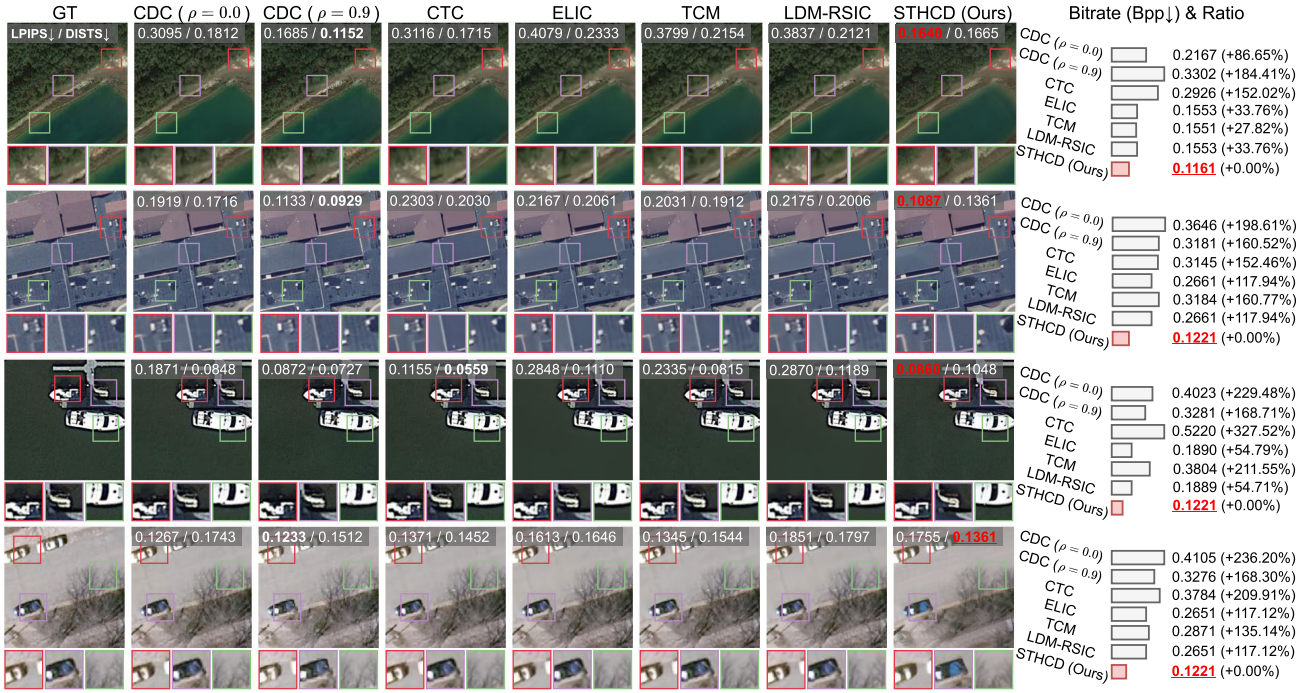


Fig. 5. Visual comparison under similar perceptual quality (Iso-quality) conditions. While the reconstructed images appear visually comparable across methods, the bitrate and ratio charts on the right highlight the significant efficiency of our method. Moreover, our approach achieves noticeably better R-D tradeoffs, delivering higher compression efficiency without sacrificing perceptual fidelity.

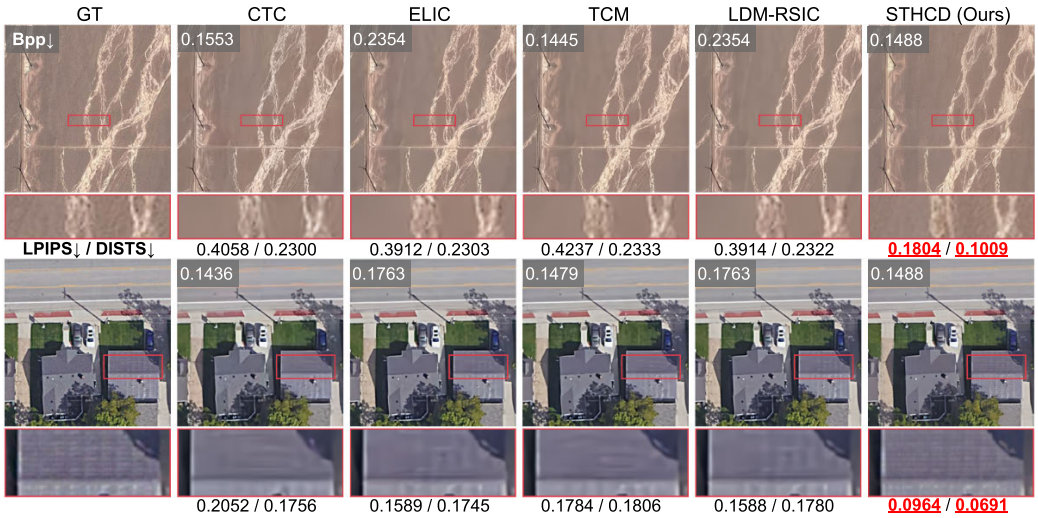


Fig. 6. Visual comparison under low bitrate constraints (Iso-bitrate, bpp $\approx$ 0.15). At this challenging bitrate, distinct visual differences emerge. Baseline LIC methods suffer from blurring and loss of fine texture, while STHCD (Ours) maintains sharp edges and realistic details, evidenced by the significantly lower LPIPS and DISTS scores (marked in red). This demonstrates the robustness of our framework in preserving structural consistency and texture details even at extremely constrained bitrates.

in red. MulimgView3[6] is used to select and zoomed-in view on specific areas of the reconstructed map.

In terms of compression efficiency, our method demonstrates significant advantages. In all displayed scenarios, STHCD completes image reconstruction with the lowest bitrate (bpp). For instance, in the building scenario (Row 2), STHCD achieves a bitrate of only 0.1221 bpp, saving over 60% of the bitstream compared to CDC ($\rho = 0.0$) at 0.3646 bpp and TCM at 0.3184 bpp, indicating superior coding performance. As illustrated in Fig. 7, STHCD consistently achieves better visual quality at lower bitrates compared to VVC and CTC (progressive coding method) on "center_56" from RSICD.

The encoding and decoding times for various LIC methods are presented in Table II. For other learned compression methods, the times correspond to the configurations achieving LPIPS $<$ 0.3 on the test dataset. For TCD and STHCD, the times are measured under the same bitrate (0.1221 bpp) and LPIPS $<$ 0.3. Since the encoding and decoding processes of TCD and STHCD are identical, their corresponding times are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.



Fig. 7. Visual comparison of reconstructed RS images at different bitrates using progressive coding methods (STHCD and CTC) and traditional codec VVC. STHCD demonstrates superior coding performance throughout the scalable encoding process.

TABLE II
ENCODING AND DECODING TIME COMPARISON ACROSS LEARNED COMPRESSION METHODS. STHCD IS EVALUATED UNDER VARIOUS TIMESTEPS

Method	Encoding Time (s)	Decoding Time (s)
CDC [23]	0.36	0.36
CTC [29]	1.23	9.95
HiFiC [10]	0.28	0.34
ELIC [6]	**0.17**	**0.13**
TCM [47]	0.30	0.27
LDM-RSIC [15]	**0.17**	0.24
TCD ($T = 1000$)	22.39	22.39
STHCD (Ours, $T = 125$)	2.80	2.80
STHCD (Ours, $T = 80$)	1.78	1.78
STHCD (Ours, $T = 15$)	0.34	0.34

also the same. This ensures a fair comparison of computational efficiency across methods under similar perceptual quality constraints.

Compared to TCD, STHCD achieves significant acceleration, reducing encoding and decoding times by up to 6.6× (e.g., at $T = 15$). Furthermore, STHCD demonstrates competitive performance relative to other learned compression methods, achieving a favorable balance between perceptual quality and computational efficiency.

D. Analysis of Spatio-Temporal Information Redundancy

To justify the effectiveness of coarse-grained representation in early low-frequency phases, we quantitatively analyze the codebook-index behavior along the diffusion trajectory. At each step t, we compute the Shannon entropy of the marginal index distribution and the spatial homogeneity measuring interimage redundancy

$$\text{Homogeneity}_t = \frac{1}{H \times W} \sum_{h,w} \max_{\mathbf{k}} \frac{\#\left\{i | \mathbf{k}_t^{(i)}[h,w] = \mathbf{k}\right\}}{N} \quad (19)$$

where $\mathbf{k}_t^{(i)}[h,w]$ is the codebook index at location (h,w) for image i at step t. High homogeneity (≈ 1.0) indicates identical indices across images at corresponding locations, exhibiting coarse, block-like structures with low spatial frequency, justifying our coarser quantization in early stages.

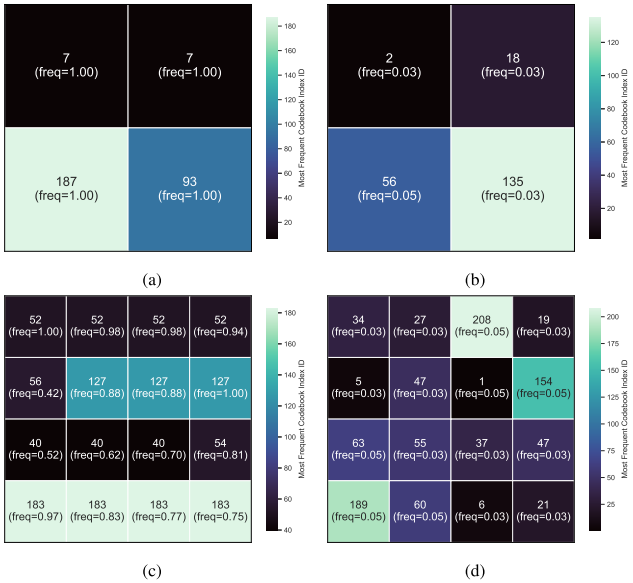


Fig. 8. Spatial distribution and frequency of codebook index selections at different diffusion steps. (a) and (b) (First row) results with unshuffle factor $s = 2$ ($T = 250$) at early ($t = 0$) and late ($t = 199$) stages, while (c) and (d) (second row) same analysis with coarser spatial partition ($s = 4$, $T = 62$) at $t = 0$ and $t = 49$, respectively. Early stages exhibit highly uniform spatial patterns with dominant mode indices, while later stages show diverse index distributions.

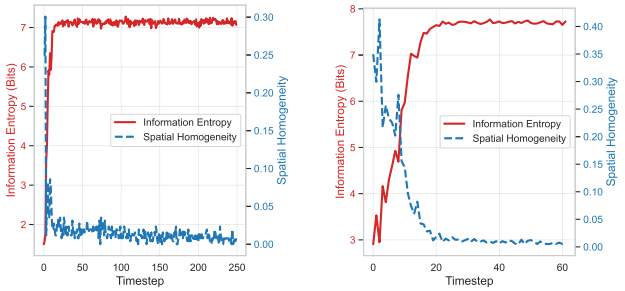


Fig. 9. Quantitative evidence of early-stage redundancy. (a) Results for $s = 2$ ($T = 250$). (b) Results for $s = 4$ ($T = 62$) both demonstrate that entropy of selected codebook indices remains low at the beginning, while spatial homogeneity is extremely high, confirming that different images select almost identical indices at corresponding spatial locations in early diffusion phases.

Fig. 8 visualizes the spatial distribution of codebook indices at different diffusion steps. For each spatial block, we show the most frequent (mode) index across all samples and its corresponding occurrence frequency. Each subfigure represents a specific diffusion timestep, where the color indicates the mode codebook index and the annotated value "freq" denotes its occurrence frequency.

At early timesteps [see Fig. 8(a) and (c)], most spatial blocks exhibit extremely high mode frequencies, indicating strong spatial correlation and intersample redundancy. As diffusion proceeds [see Fig. 8(b) and (d)], the index distributions become increasingly diverse, motivating a CTF coding strategy that allocates lower bitrate to early stages and finer encoding to later stages.

Fig. 9 provides quantitative evidence: the entropy of selected codebook indices remains low at the beginning, while spatial homogeneity is extremely high, empirically proving that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

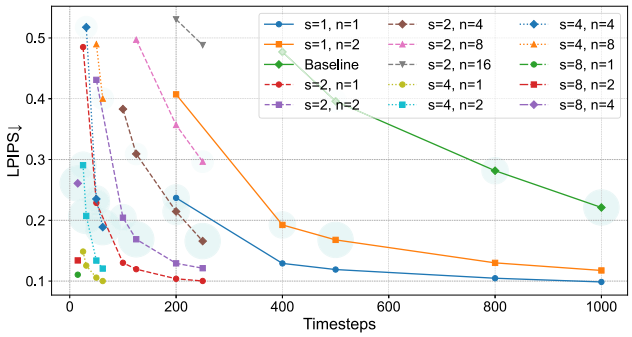


Fig. 10. Ablation study on the residual transform factor s and codebook splitting factor n. Light blue shaded circles with same size indicate data points with equivalent bitrates.

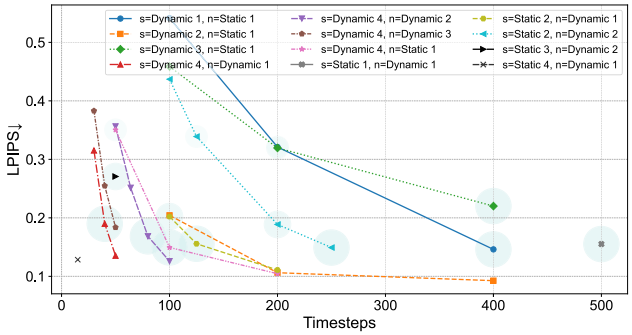


Fig. 11. Ablation study on static and dynamic residual transform and codebook splitting. Light blue shaded circles with same size indicate data points with equivalent bitrates.

fine-grained spatial representation is unnecessary in early phases because coding decisions are dominated by deterministic, low-frequency structural priors.

E. Ablation Study

As shown in Fig. 10, we conduct an ablation study on the residual transform factor (s) and the codebook splitting factor (n). Using $s = 1$, $n = 1$ as the baseline (green diamond line), significant performance improvements were observed.

The residual transform accelerates the convergence of the diffusion model. Introducing $s > 1$ reduces the required timesteps to achieve similar perceptual quality (measured by LPIPS). For example, the baseline requires 400 timesteps to reach an LPIPS score of 0.13, while $s = 2$, $n = 1$ (red dashed line) achieves the same score in about 100 timesteps, a 4× speedup. Increasing s to 8 (green dotted line) achieves comparable quality to the baseline at 800 timesteps in fewer than 50 timesteps.

The proposed method also improves reconstruction quality at the same bitrate. Light blue circles in the figure indicate data points with equivalent bitrates. Higher s values shift these points toward the lower-left region, showing reduced LPIPS error and inference latency under the same bitrate, demonstrating the effectiveness of the residual transform and codebook splitting.

Fig. 11 presents an ablation study analyzing the trade-off between perceptual quality (LPIPS) and inference speed (timesteps) across different configurations of residual transform shuffle coefficient s and codebook splitting channel width n. For static s, labels Static 1–4 correspond to fixed shuffle factors of 1, 2, 4, and 8, respectively. Dynamic s variants adaptively mix these factors: Dynamic 1 and 3 combine factors 1 and 2, Dynamic 2 combines 1 and 4, and Dynamic 4 combines 2 and 4. For codebook splitting n, Static 1 uses fixed 4-channel configuration, while Dynamic 1–3 employ mixed channel widths (Dynamic 1:4, 2, 1 channels; Dynamic 2:8, 4, 2 channels; Dynamic 3:4, 2 channels). The results demonstrate that dynamic strategies significantly accelerate generation. Notably, s = Dynamic 4 with n = Dynamic 1 achieves convergence within 30–50 timesteps, compared to 200–400 timesteps for s = Dynamic 1 or static settings, while maintaining competitive LPIPS scores. This confirms that the proposed dynamic mechanisms successfully balance computational efficiency and image quality.

TABLE III
ABLATION STUDY OF RESIDUAL TRANSFORM AND CODEBOOK SPLITTING

Residual Transform	Codebook Splitting	PSNR ↑	LPIPS ↓	DISTS ↓	FID ↓	Steps ↓
✗	✗	25.66	0.2214	0.1751	96.21	1000
✗	✓	26.10	0.1683	0.1469	82.36	500
✓	✗	**26.15**	**0.1662**	**0.1455**	**80.21**	**250**

The ablation study in Table III evaluates the impact of two key components: Residual Transform and Codebook Splitting. The results demonstrate that each component contributes to improved performance, with the full configuration (all components enabled) achieving the best tradeoff between reconstruction quality (PSNR, LPIPS, DISTS and FID) and computational efficiency (Timesteps). To ensure a fair comparison, the encoding bitrate is kept consistent across all configurations (bpp = 0.1221), allowing the study to focus on differences in reconstruction quality and encoding time.

To verify that the residual unshuffle operation does not introduce block artifacts or disrupt linear structures, we conduct visual analysis of structural preservation through edge maps and error maps. Fig. 12 presents Canny edge maps at matched bitrates, demonstrating that salient edges and long straight structures remain intact under residual unshuffle. Critically, we observe no emergence of spurious block-like edges or artificial straight boundaries; edge localization and continuity are preserved across diverse examples containing both textured regions and line-structured content.

To validate the effectiveness of the proposed CTF strategy, we compare it against a reverse fine-to-coarse (FTC) schedule. Fig. 13 shows reconstructed images and their absolute error maps relative to GT (brighter pixels indicate larger distortion). The CTF approach achieves consistently lower perceptual distortion (LPIPS/DISTS) and better preserves high-frequency textures. In contrast, FTC allocates coarse representations in later diffusion stages, which impairs the progressive refinement of detailed structures and leads to larger residual errors. This validates that coding granularity should align with the intrinsic CTF information hierarchy of the diffusion generation process.

Table IV provides a detailed comparison of the codebook storage size and the corresponding storage improvement under

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

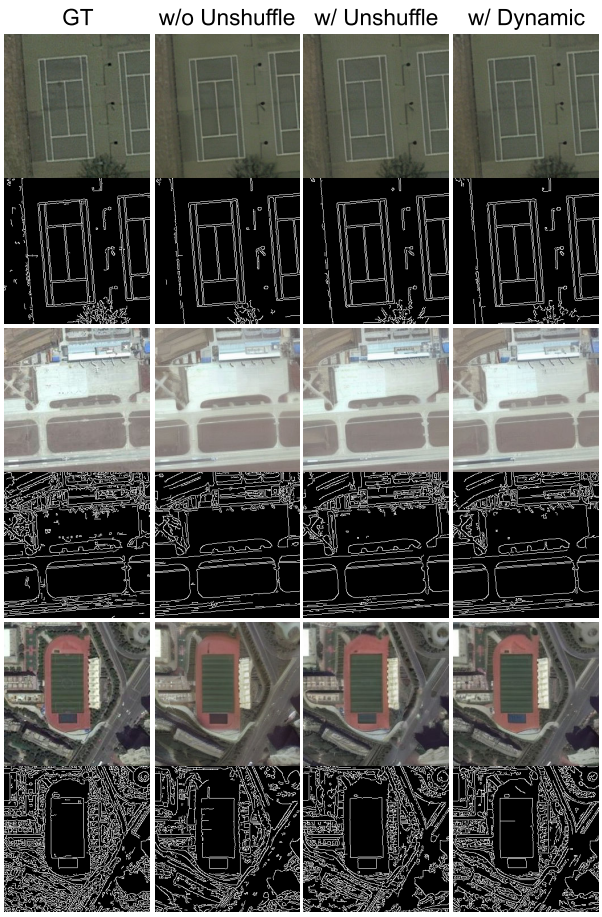


Fig. 12. Reconstructed images with Canny edge maps. The residual unshuffle preserves major edges and linear structures without introducing spurious block-like boundaries.

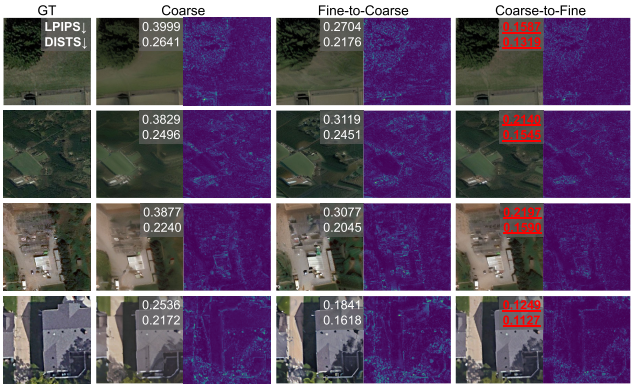


Fig. 13. Visual comparison of CTF and FTC diffusion coding strategies at comparable bitrate (bpp ≈ 0.12). Error maps (brighter = larger distortion) show that CTF better preserves structural details.

various configurations of timesteps (T), residual transform scaling factor (s), and codebook splitting factor (n). The first section of the table illustrates the impact of reducing the number of timesteps while keeping s and n fixed. As the number of timesteps decreases from 1000 to 15, the storage size is significantly reduced, achieving up to a 98.50% improvement. The second section evaluates the effect of varying the residual transform scaling factor s while maintaining $T = 1000$ and $n = 4$. Increasing s results in a reduction of spatial dimensions (H and W), leading to storage reduction ranging from 75.00% to 93.75%. The third section explores the influence of the codebook splitting factor n while keeping $T = 1000$ and $s = 1$. As n decreases, the storage size is reduced, with improvements of 50.00% and 75.00% for $n = 2$ and $n = 1$, respectively. Finally, the generalized formula in the last row of the table provides a unified representation of the storage reduction as a function of T, s and n. This formula highlights the tradeoffs between these parameters and their combined effect on storage efficiency. The results demonstrate the flexibility of the proposed approach in optimizing storage requirements under different configurations.

TABLE IV

CODEBOOK STORAGE SIZE COMPARISON WITH DIFFERENT CONFIGURATIONS

Timestep	s	n	Codebook Dimension	Storage Reduction
1000	1	4	$[1000, K, H, W, 4]$	Baseline
125	1	4	$[125, K, H, W, 4]$	87.50%
15	1	4	$[15, K, H, W, 4]$	**98.50%**
1000	2	4	$[1000, K, H//2, W//2, 4]$	75.00%
1000	4	4	$[1000, K, H//4, W//4, 4]$	93.75%
1000	1	2	$[1000, K, H, W, 2]$	50.00%
1000	1	1	$[1000, K, H, W, 1]$	75.00%
T	s	n	$[T, K, H//s, W//s, n]$	$(1 - \frac{T}{1000} \cdot \frac{1}{s^2} \cdot \frac{n}{4}) \times 100\%$

TABLE V

R-D PERFORMANCE COMPARISON ON SAR IMAGES (DSSDD TEST SET). STHCD ACHIEVES BETTER PERCEPTUAL QUALITY (LOWER LPIPS/DISTS) AT A LOWER BITRATE COMPARED TO JPEG2000

Method	LPIPS ↓	DISTS ↓	bpp ↓
JPEG2000	0.6408	0.3361	0.1827
STHCD (Ours)	**0.1540**	**0.1635**	**0.1514**

F. Cross-Modality Generalization

To assess the generalization capabilities of STHCD, we extended it to synthetic aperture radar (SAR) and HSI using lightweight preprocessing without model fine-tuning.

1) SAR: We utilize the dual-polarimetric SAR ship detection dataset (DSSDD) [52]. Since standard SAR data are complex-valued with high dynamic range, we transform the dual-polarimetric covariance matrix

$$C_2 = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \tag{20}$$

into a three-channel format by extracting amplitudes: $R = |C_{11}|$, $G = |C_{12}|$, $B = |C_{22}|$, where C_{11}, C_{22} are copolarized components and C_{12} encodes cross-polarization. This preserves scattering characteristics while enabling RGB-based processing.

We compare our STHCD against JPEG2000 at similar bitrates on the DSSDD test set. As shown in Table V, while JPEG2000 achieves a higher PSNR (favoring pixel-wise fidelity), our STHCD achieves significantly lower LPIPS and DISTS scores. This indicates that our generative approach better preserves the high-frequency textures and scattering patterns of SAR images, which are perceptually more aligned with the original data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

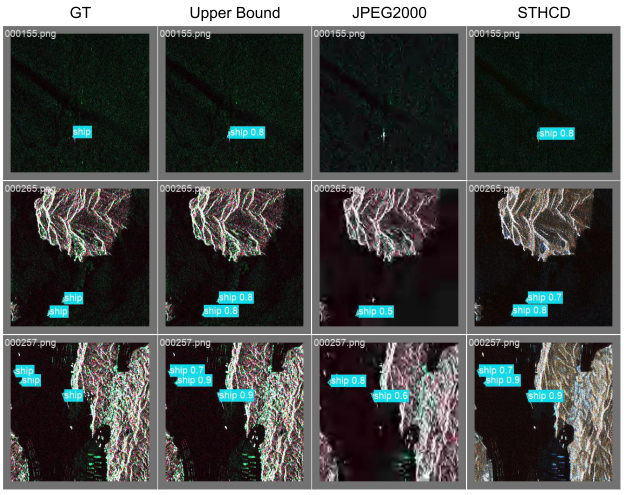


Fig. 14. Visual comparison of ship detection results on the DSSDD test set. Upper Bound stands for test results on uncompressed images. JPEG2000 with average bpp = 0.1826 and STHCD with average bpp = 0.1514. Our method preserves ship structures more effectively, reducing false negatives and maintaining high confidence scores.

TABLE VI

QUANTITATIVE COMPARISON BETWEEN STHCD AND JPEG2000 ON IP DATASET USING BAND GROUPING AND PCA STRATEGIES. STHCD CONSISTENTLY ACHIEVES BETTER SPECTRAL FIDELITY AND PERCEPTUAL QUALITY AT LOWER BITRATES

Method	SAM ↓	LPIPS ↓	DISTS ↓	bpppb ↓
Band Grouping				
JPEG2000	2.9052	0.2106	0.2101	0.0610
STHCD (Ours)	**2.8584**	**0.0936**	**0.1137**	**0.0505**
PCA with 3 components				
PCA Upper Bound	*1.9459*	–	–	–
JPEG2000	3.1533	0.2460	0.2444	0.0009
STHCD (Ours)	**2.9537**	**0.1224**	**0.1525**	**0.0008**
PCA with 9 components				
PCA Upper Bound	*1.3085*	–	–	–
JPEG2000	2.9631	0.3906	0.2705	0.0027
STHCD (Ours)	**2.7654**	**0.1767**	**0.1670**	**0.0023**
PCA with 30 components				
PCA Upper Bound	*0.6135*	–	–	–
JPEG2000	2.8881	0.4378	0.2791	0.0274
STHCD (Ours)	**2.7477**	**0.1977**	**0.1695**	**0.0226**

Results demonstrate that JPEG2000 suffers from severe blurring at low bitrates, leading to confidence drops (visualized in Fig. 14). In contrast, our STHCD preserves the structural integrity of ships better.

2) Hyperspectral Imagery: We utilize the Indian Pines (IP) dataset. To adapt the 200-band data to the 3-channel VAE interface, we employ two strategies: 1) *band grouping:* where bands are split into groups of three and 2) *PCA reduction:* where the top $n \in \{3, 9, 30\}$ principal components are extracted (lossy representation).

We use bpp per band (bpppb) as the bitrate metric. For spectral fidelity, we verify the spectral angle mapper (SAM) [53] and downstream Land Cover Classification accuracy overall accuracy (OA).

Table VI summarizes the results. In the Band Grouping setting, STHCD achieves better spectral fidelity (lower SAM 2.8584 versus 2.9052) and perceptual metrics than JPEG2000.

TABLE VII

CLASSIFICATION PERFORMANCE ON IP DATASET IN BAND GROUPING SETTING. "UPPER BOUND" REPRESENTS THE RESULTS OBTAINED BY DIRECTLY TESTING ON THE ORIGINAL IMAGES AFTER PREPROCESSING AND RESTORATION

Method	OA ↑	AA ↑	Kappa ↑
Band Grouping			
Upper Bound	*0.9728*	*0.9762*	*0.9787*
JPEG2000	0.6981	0.7375	0.8103
STHCD (Ours)	**0.7094**	**0.7506**	**0.8498**
PCA with 3 components			
Upper Bound	*0.4550*	*0.5100*	*0.3601*
JPEG2000	0.4132	**0.4331**	0.3003
STHCD (Ours)	**0.4322**	0.3966	**0.3192**
PCA with 9 components			
Upper Bound	*0.6417*	*0.8011*	*0.5872*
JPEG2000	0.4973	0.5583	0.4041
STHCD (Ours)	**0.5085**	**0.6169**	**0.4243**
PCA with 30 components			
Upper Bound	*0.8041*	*0.9016*	*0.7747*
JPEG2000	0.6043	0.6750	0.5310
STHCD (Ours)	**0.6762**	**0.7861**	**0.6182**

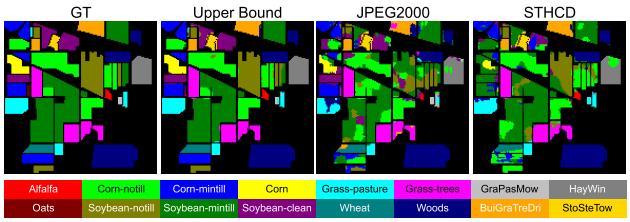


Fig. 15. Visual comparison of classification maps using Band Grouping strategy. From (left to right): GT, upper bound (result on uncompressed data), JPEG2000, and the proposed STHCD. The proposed method preserves clearer class boundaries and reduces misclassification artifacts compared to JPEG2000.

Crucially, for the downstream classification task, STHCD consistently yields higher OA across both strategies. Even with aggressive PCA reduction ($n = 9$), STHCD maintains high classification accuracy (e.g., mention specific number here if not in table), verifying that task-relevant semantics are preserved.

To validate the effectiveness of compressed representations for downstream analysis, Table VII presents classification performance using DBDA [54] classifier. In the Band Grouping setting, STHCD (OA: 0.8498) significantly outperforms JPEG2000 (OA: 0.8103) while achieving lower bitrate, demonstrating superior preservation of discriminative features. For PCA-based compression, both methods show performance degradation compared to the original data, as PCA reconstruction inherently introduces information loss. However, STHCD consistently maintains a smaller performance gap across all component settings. With $n = 30$ components, STHCD achieves OA of 0.6762 versus JPEG2000's 0.6043, reducing the accuracy drop by 40% relative to JPEG2000's degradation. This validates that STHCD better retains task-relevant semantic information even under aggressive dimensionality reduction.

The qualitative classification maps in Fig. 15 (Band Grouping) reveal that STHCD preserves boundary distinctness and class consistency.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
TECHNICAL SPECIFICATIONS OF SIMULATED GROUND STATION AND ON-BOARD EMBEDDED SYSTEM

System Role	Ground Station	LEO Satellite
GPU Name	GeForce RTX 4090	Jetson Orin NX
Architecture	Ada Lovelace	Ampere
Compute Performance	1321 AI TOPs	100 (S) INT8 TOPs
CUDA Cores	16384	1024
Tensor Cores	512	32
Memory	24 GB GDDR6X	16 GB LPDDR5
Memory Interface	384-bit	128-bit
TGP/Power	450 W	**10-25 W**

TABLE IX
ENCODING/DECODING TIME COMPARISON WITH AND WITHOUT TENSORRT ACCELERATION. TCD CORRESPONDS TO $T = 1000$, AND STHCD IS EVALUATED UNDER VARIOUS TIMESTEPS ($T = 125, 100, 80, 15$)

Method	Timesteps (T)	w/o TensorRT (s)	w/ TensorRT (s)
TCD (4090)	1000	22.36	3.82
STHCD (Ours, 4090)	125	2.80	0.48
	100	2.25	0.39
	80	1.78	0.31
	15	**0.34**	**0.06**
STHCD (Ours, Jetson)	125	38.36	6.49
	100	30.57	5.27
	80	24.48	4.21
	15	**4.60**	**0.81**

G. On-Board Deployment Simulation

Efficient RSIC is critical for on-board deployment across diverse computational resources. To evaluate STHCD's practical applicability, we conduct experiments on two platforms, as shown in Table VIII: an RTX 4090 GPU simulating a ground station (1321 AI TOPs, 24 GB memory) and a Jetson Orin NX embedded system simulating a LEO satellite (100 INT8 TOPs, 10–25 W). This dual-platform evaluation demonstrates STHCD's versatility across high-throughput ground processing and energy-efficient on-board scenarios. TensorRT was employed to further accelerate inference.

The encoding and decoding times of STHCD and TCD under various timesteps, with and without TensorRT acceleration, are presented in Table IX. The results show that TensorRT significantly reduces computational time, achieving up to a 5.85× speedup. For instance, at $T = 15$, the encoding/decoding time of STHCD decreases from 0.34 to 0.06 s with TensorRT acceleration. This demonstrates the efficiency of STHCD in both high-performance and resource-constrained scenarios. Furthermore, the comparison highlights that STHCD achieves competitive performance relative to TCD while offering substantial acceleration, especially at lower timesteps.

In summary, the proposed STHCD framework is not only computationally efficient but also highly adaptable to different hardware platforms, making it a robust solution for RSIC in both ground station and on-board satellite environments.

V. DISCUSSION AND LIMITATIONS

The proposed STHCD framework introduces a novel approach to LIC for RS by leveraging hierarchical diffusion models to optimize resource allocation. Through spatial residual transformations, channel-wise codebook splitting, and a time-dependent hierarchical policy, STHCD effectively balances perceptual quality and computational efficiency.

Although the diffusion backbone is pretrained on natural RGB imagery, the coding pipeline itself is modality-agnostic. Our results on SAR (DSSDD) and HSI (IP) demonstrate that when non-RGB modalities are mapped to a compatible 3-channel representation via intensity/amplitude channels for SAR or band grouping/PCA for HSI—the generative prior effectively: 1) reconstructs spatial textures and structural patterns and 2) preserves task-relevant semantics for downstream models (e.g., detection and classification). Crucially, this extension remains training-free and requires only lightweight preprocessing, making it practical when modality-specific diffusion models are unavailable.

However, several limitations arise from the current training-free implementation using a fixed RGB-pretrained backbone:

A. Channel Mismatch

The 3-channel VAE interface constrains nonoptical data representation. For SAR, mapping complex-valued measurements to amplitude/intensity channels preserves structural cues and detection accuracy (mAP@50: 0.9870 versus 0.9810 for JPEG2000) but discards phase information critical for InSAR applications. For HSI with 200 bands, dimensionality reduction is necessary—PCA introduces irreversible information loss, while band grouping may underutilize spectral correlations.

B. Domain Gap

The distribution mismatch between nonoptical modalities (with heavy-tailed statistics) and natural RGB images limits optimal utilization of the generative prior, particularly in extreme-value regions. While STHCD outperforms JPEG2000 at low bitrates, this gap suggests potential for further downstream task improvements.

C. Computational Costs

While training-free, diffusion inference incurs higher computational costs than conventional codecs. This can be mitigated through standard acceleration techniques (fewer denoising steps, fast samplers, distillation) to achieve deployment-friendly latency-quality tradeoffs.

Importantly, these limitations are backbone-specific rather than fundamental to STHCD. The coding pipeline is modality-agnostic and can readily adopt future domain-specific diffusion models (e.g., SAR-native or multichannel backbones) to overcome these constraints while preserving the entropy modeling and hierarchical allocation mechanisms.

VI. CONCLUSION

This article addresses the challenges of LIC in RS, where generative models, particularly diffusion-based methods, have shown great potential but face limitations such as hallucination of fine details at low bitrates and inefficiency due to long

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

inference times. We propose STHCD, a training-free framework that aligns resource allocation with the CTF hierarchy of diffusion models. STHCD features: 1) spatial residual transformation via unshuffle operations; 2) channel-wise codebook splitting for flexible bit allocation; and 3) a time-dependent hierarchical policy to coordinate operations across timesteps. Extensive experiments demonstrate that STHCD significantly outperforms state-of-the-art methods in perceptual metrics. At equivalent bitrates, STHCD achieves a significant speedup over standard TCD while optimizing perceptual quality, establishing a new paradigm for hierarchical RS compression.

Although the current implementation uses an RGB-pretrained diffusion backbone, the coding pipeline itself is modality-agnostic. Experiments on SAR and HSI data demonstrate that simple channel mapping (amplitude/intensity for SAR, band grouping/PCA for HSI) enables effective compression with preserved downstream task performance. However, constraints such as 3-channel VAE interface, domain gap between nonoptical and natural images, and diffusion inference costs remain. Importantly, these are backbone-specific limitations rather than fundamental to STHCD—the framework can readily adopt future domain-specific diffusion models while preserving its hierarchical entropy modeling and allocation mechanisms.

Looking ahead, the flexibility of STHCD opens new possibilities for adapting to diverse operational requirements. Future work will explore lightweight policy networks for automatic phase boundary learning and per-step resource allocation, further enhancing the framework's adaptability and efficiency. Additionally, future research will focus on extending the framework to handle multimodal RS data with domain-specific backbones, broadening its applicability across various domains.

REFERENCES

[1] M. Burke, A. Driscoll, D. B. Lobell, and S. Ermon, "Using satellite imagery to understand and promote sustainable development," *Science*, vol. 371, no. 6535, pp. 1–13, 2021.

[2] Y. Li, Q. Hou, Z. Zheng, M.-M. Cheng, J. Yang, and X. Li, "Large selective kernel network for remote sensing object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2023, pp. 16794–16805.

[3] F. Bellard. (2017). *BPG Image Format*. Accessed: Jan. 26, 2026. [Online]. Available: http://bellard.org/bpg/

[4] B. Bross et al., "Overview of the versatile video coding (VVC) standard and its applications," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 10, pp. 3736–3764, Oct. 2021.

[5] J. Ballé, V. Laparra, and E. Simoncelli, "End-to-end optimized image compression," in *Proc. 5th Int. Conf. Learn. Rep. (ICLR)*, Toulon, France, Apr. 2017, pp. 1–27.

[6] D. He, Z. Yang, W. Peng, R. Ma, H. Qin, and Y. Wang, "ELIC: Efficient learned image compression with unevenly grouped space-channel contextual adaptive coding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5718–5727.

[7] A. Van Den Oord, O. Vinyals, and K. Kavukcuoglu, "Neural discrete representation learning," in *Proc. Adv. Neural Inf. Process. Syst. Annu. Conf. Neural Inf. Process. Syst.*, Long Beach, CA, USA, Dec. 2017, pp. 6306–6315.

[8] A. Razavi, A. van den Oord, and O. Vinyals, "Generating diverse high-fidelity images with VQ-VAE-2," in *Proc. 33rd Int. Conf. Neural Inf. Process. Syst.*, 2019, pp. 14866–14876.

[9] G. Ohayon, H. Manor, T. Michaeli, and M. Elad, "Compressed image generation with denoising diffusion codebook models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2025, pp. 1–41.

[10] F. Mentzer, G. D. Toderici, M. Tschannen, and E. Agustsson, "High-fidelity generative image compression," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 11913–11924.

[11] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[12] P. Han, B. Zhao, and X. Li, "Edge-guided remote-sensing image compression," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524515.

[13] J. Zhang et al., "Spatial–temporal context model for remote sensing imagery compression," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 6063–6072.

[14] J. Li and X. Hou, "Object-fidelity remote sensing image compression with content-weighted bitrate allocation and patch-based local attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 2004314.

[15] J. Li, J. Li, X. Hou, and H. Wang, "Exploring distortion prior with latent diffusion models for remote sensing image compression," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5623713.

[16] Z. Zhang et al., "COSMIC: Compress satellite images efficiently via diffusion compensation," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024, pp. 10–15.

[17] J. Yang, B. Du, and L. Zhang, "From center to surrounding: An interactive learning framework for hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 197, pp. 145–166, Mar. 2023.

[18] F. Luo et al., "SCMVC: Semantic constraint-based spatial–spectral multiview clustering for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5533013.

[19] R. Liu et al., "DHSNet: Dual classification head self-training network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5534515.

[20] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 6840–6851.

[21] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 1–20.

[22] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.

[23] R. Yang and S. Mandt, "Lossy image compression with conditional diffusion models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 36, 2023, pp. 64971–64995.

[24] Y. Cheng, F. Meng, Z. Zhang, R. Ding, Y. Wang, and Y. Liang, "Progressive diffusion-based low rate perceptual image compression with discrete Gaussian codebooks for remote sensing," in *Proc. Chin. Conf. Pattern Recognit. Comput. Vis. (PRCV)*. Cham, Switzerland: Springer, 2025, pp. 138–152.

[25] G. K. Wallace, "The JPEG still picture compression standard," *Commun. ACM*, vol. 34, no. 4, pp. 30–44, Apr. 1991.

[26] A. Skodras, C. Christopoulos, and T. Ebrahimi, "The JPEG 2000 still image compression standard," *IEEE Signal Process. Mag.*, vol. 18, no. 5, pp. 36–58, Sep. 2001.

[27] Y. Lu, Y. Zhu, Y. Yang, A. Said, and T. S. Cohen, "Progressive neural image compression with nested quantization and latent ordering," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 539–543.

[28] J.-H. Lee, S. Jeon, K. P. Choi, Y. Park, and C.-S. Kim, "DPICT: Deep progressive image compression using trit-planes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16092–16101.

[29] S. Jeon, K. P. Choi, Y. Park, and C.-S. Kim, "Context-based trit-plane coding for progressive image compression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14348–14357.

[30] T. Li, C. Li, S. Yin, Y. Bao, F. Meng, and Y. Liang, "Fine-grained adjustable entropy models for rate-complexity jointly adjustable image compression," in *Proc. Chin. Conf. Pattern Recognit. Comput. Vis. (PRCV)*. Cham, Switzerland: Springer, 2024, pp. 3–14.

[31] C. Li et al., "One is all: A unified rate-distortion-complexity framework for learned image compression under energy concentration criteria," *IEEE Trans. Multimedia*, vol. 27, pp. 3992–4007, 2025.

[32] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018.

[33] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, Nov. 2010, pp. 270–279.

[34] C. Wang and W. Sun, "Controllable reference guided diffusion with local global fusion for real world remote sensing image super resolution," 2025, *arXiv:2506.23801*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.

[35] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[36] J. Ding, M. Elhoseiny, and X. Li, "VRSBench: A versatile vision-language benchmark dataset for remote sensing image understanding," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 3229–3242.
[37] J. Ding et al., "Object detection in aerial images: A large-scale benchmark and challenges," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7778–7796, Nov. 2022.
[38] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.
[39] G. Christie, N. Fendley, J. Wilson, and R. Mukherjee, "Functional map of the world," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6172–6180.
[40] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.
[41] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.
[42] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.
[43] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 586–595.
[44] K. Ding, K. Ma, S. Wang, and E. P. Simoncelli, "Image quality assessment: Unifying structure and texture similarity," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2567–2581, May 2022.
[45] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst. (NIPS)*, 2017, pp. 1–12.
[46] (2015). *WebP Compression Study*. Accessed: Jan. 26, 2026. [Online]. Available: https://developers.google.com/speed/webp/docs/webp_study
[47] J. Liu, H. Sun, and J. Katto, "Learned image compression with mixed transformer-CNN architectures," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14388–14397.
[48] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.
[49] M. Everingham, S. M. A. Eslami, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes challenge: A retrospective," *Int. J. Comput. Vis.*, vol. 111, no. 1, pp. 98–136, Jan. 2015.
[50] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.
[51] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proc. 23rd Int. Conf. Mach. Learn.*, 2006, pp. 233–240.
[52] Y. Hu, Y. Li, and Z. Pan, "A dual-polarimetric SAR ship detection dataset and a memory-augmented autoencoder-based detection method," *Sensors*, vol. 21, no. 24, p. 8478, Dec. 2021.
[53] R. H. Yuhas, A. F. H. Goetz, and J. W. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm," in *Proc. 3rd Annu. JPL Airborne Geosci. Workshop*, Pasadena, CA, USA, 1992, pp. 147–149.
[54] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.



Yangxuan Cheng received the B.S. degree in computer science and technology from Harbin Institute of Technology, Weihai, China, in 2024. He is currently pursuing the Ph.D. degree with the School of Information Science and Technology, Harbin Institute of Technology, Shenzhen, China.

His research interests include image compression, deep learning, and intelligent communication.



Fanyang Meng is currently an Associate Research Fellow with the Pengcheng Laboratory, Shenzhen, China. His research interests include video coding, computer vision, human action recognition, as well as network coding and semantic communication.



Hao Qi received the M.S. degree in computer science from the Ocean University of China, Qingdao, China, in 2024. He is currently pursuing the Ph.D. degree with the School of Information Science and Technology, Harbin Institute of Technology, Shenzhen, China.

His research interests include computer vision, image enhancement, image compression, and video compression.



Han Shen received the B.E. degree in applied physics from Yanshan University, Qinhuangdao, China, in 2020, and the M.Eng. degree in electrical science and technology from Harbin Institute of Technology, Harbin, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Information Science and Technology, Harbin Institute of Technology, Shenzhen, China.

His research interests include network coding and deep learning applications in wireless communication.



Zhongqiang Zhang received the Ph.D. degree from the School of Artificial Intelligence, Xidian University, Xi'an, China, in 2021.

Since 2021, he has been with the Pengcheng Laboratory, Shenzhen, China, as an Assistant Researcher. His research interests include semantic communication and its applications in satellite earth and underwater scenarios.



Ye Wang (Member, IEEE) received the M.S. and Ph.D. degrees in information and communication engineering from Harbin Institute of Technology, Harbin, China, in 2009 and 2013, respectively.

From 2013 to 2014, he was a Post-Doctoral Research Fellow with the University of Ontario Institute of Technology, Oshawa, ON, Canada. From 2015 to 2021, he was an Assistant Professor with Harbin Institute of Technology, Shenzhen, China. Since 2022, he has been an Associate Professor with the Pengcheng Laboratory, Shenzhen. His research interests include satellite communications, resource allocation, and mobile internet.



Yongsheng Liang (Member, IEEE) received the Ph.D. degree in mechanical engineering from Harbin Institute of Technology, Harbin, China, in 1999.

From 2002 to 2005, he was a Post-Doctoral Researcher of information and communication engineering at Harbin Institute of Technology. Since 2017, he has been a Professor with the School of Electronics and Information Engineering, Harbin Institute of Technology, Shenzhen. He is currently the Chair Professor with Shenzhen Technology University, Shenzhen, and a Distinguished Professor with Shenzhen University, Shenzhen. He is the author of two books, more than 70 articles, and more than ten invention patents. His research interests include video coding and transferring, neural networks, and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:06:36 UTC from IEEE Xplore. Restrictions apply.